

A056 v0.2.3

PKLSA Spectral–Sine–Gordon–Mittag–Leffler Falsifier

SST-Workbench

Framework v1.0.4 CANONICAL_FROZEN

1 Identification

Catalog A056, release v0.2.3. This falsifier is a thin scientific instance of the immutable SST Falsifier Framework v1.0.4. The framework is referenced through `.sst_framework_root`; it is not vendored or patched by this release.

2 Question

Does provider-native vortex-filament phase dynamics require an effective Sine–Gordon restoring law and/or Mittag–Leffler fractional-memory relaxation after held-out confirmation, numerical certification and evidence-class-aware replication?

3 Hypotheses

The phase null family consists of Wave, Klein–Gordon and Duffing laws. The phase candidate is Sine–Gordon. The relaxation null family consists of exponential, stretched-exponential and bi-exponential laws; Mittag–Leffler is the fractional-memory candidate. A physical conclusion additionally requires canonical-framework eligible cross-source replication.

4 Assumptions and interpretation boundary

The phase observable is defined upstream before scoring. Time and arclength-like coordinates are monotone and approximately uniform. Python/NumPy FP64 is authoritative reference arithmetic; C++/OpenMP FP64 certifies the A056 kernels. SYCL FP32 and DD32/FP32x2 remain screening-only. In conservative inviscid dynamics, projected ringdown is not automatically interpreted as thermodynamic dissipation.

5 Symbols

Symbol	Meaning	Units
$\varphi(s, t)$	provider-declared phase/torsion observable	rad / dimensionless
a	phase-wave coefficient	$L^2 T^{-2}$
b, c	restoring coefficients	T^{-2} for dimensionless phase
α	Mittag–Leffler fractional order	1
τ	relaxation timescale	T

6 Equations

6.1 Phase family

$$H_W : \varphi_{tt} = a\varphi_{ss}, \quad (1)$$

$$H_{KG} : \varphi_{tt} = a\varphi_{ss} - b\varphi, \quad (2)$$

$$H_D : \varphi_{tt} = a\varphi_{ss} - b\varphi - c\varphi^3, \quad (3)$$

$$H_{SG} : \varphi_{tt} = a\varphi_{ss} - b \sin \varphi. \quad (4)$$

For dimensionless phase, $[a] = L^2 T^{-2}$ and $[b] = [c] = T^{-2}$.

6.2 Relaxation family

$$R_{\text{exp}}(t) = e^{-t/\tau}, \quad (5)$$

$$R_{SE}(t) = e^{-(t/\tau)^\beta}, \quad (6)$$

$$R_{2e}(t) = we^{-t/\tau_1} + (1-w)e^{-t/\tau_2}, \quad (7)$$

$$R_{ML}(t) = E_{\alpha,1}[-(t/\tau)^\alpha], \quad (8)$$

$$E_{\alpha,\beta}(z) = \sum_{k=0}^{\infty} \frac{z^k}{\Gamma(\alpha k + \beta)}. \quad (9)$$

The exact null limit $E_{1,1}(-t/\tau) = e^{-t/\tau}$ is explicitly protected by the registered interior support interval $0.48 < \alpha < 0.97$.

7 Algorithm

The pipeline validates paired NPZ/JSON provider cases, computes centered POD/SVD qualification, forms second-order temporal and spatial phase derivatives, fits all registered model families on discovery blocks, evaluates untouched confirmation blocks, performs deterministic twofold coarsening, and certifies the experiment-specific kernels against C++/OpenMP FP64. Replication is then assessed with the frozen framework evidence classes.

Machine-readable science contract

Research question Do provider-native vortex-filament phase dynamics require an effective Sine-Gordon restoring law and/or Mittag-Leffler fractional-memory relaxation after held-out confirmation, numerical certification and evidence-class-aware replication?

Objective Blindly compare preregistered phase and relaxation model families on provider-native dynamic fields while separating implementation controls, simulations and physically independent evidence.

H_0 H0: Sine-Gordon and Mittag-Leffler descriptions do not achieve the preregistered discovery, held-out, numerical-certification and independent-replication thresholds beyond the registered alternatives.

H_1 H1: one or both candidate mechanisms outperform the registered alternatives, survive held-out and numerical certification, and replicate across eligible independent physical evidence.

Registered equations

E1: Wave null phase model

$$\varphi_{tt} = a\varphi_{ss} \quad (10)$$

Dimensional check: Both sides have dimensions T^{-2} for dimensionless phase when $[a] = L^2 T^{-2}$.

E2: Klein-Gordon competitor

$$\varphi_{tt} = a\varphi_{ss} - b\varphi \quad (11)$$

Dimensional check: $[b]=T^{-2}$; every term has dimensions T^{-2} .

E3: Duffing competitor

$$\varphi_{tt} = a\varphi_{ss} - b\varphi - c\varphi^3 \quad (12)$$

Dimensional check: For dimensionless phase, $[b]=[c]=T^{-2}$.

E4: Sine-Gordon candidate

$$\varphi_{tt} = a\varphi_{ss} - b\sin\varphi \quad (13)$$

Dimensional check: $\sin(\varphi)$ is dimensionless and $[b]=T^{-2}$.

E5: Bayesian information criterion used on discovery blocks

$$\text{BIC} = n \ln(\text{RSS}/n) + k \ln n \quad (14)$$

Dimensional check: Model residuals are compared within one observable/scaling; BIC differences are dimensionless.

E6: Held-out normalized root-mean-square error

$$\text{NRMSE} = \sqrt{n^{-1} \sum_i (y_i - \hat{y}_i)^2 / \sigma_y} \quad (15)$$

Dimensional check: Numerator and denominator have the same observable units, so NRMSE is dimensionless.

E7: Mittag-Leffler function

$$E_{\alpha,\beta}(z) = \sum_{k=0}^{\infty} z^k / \Gamma(\alpha k + \beta) \quad (16)$$

Dimensional check: z must be dimensionless.

E8: Mittag-Leffler relaxation candidate

$$R_{ML}(t) = E_{\alpha,1}[-(t/\tau)^\alpha] \quad (17)$$

Dimensional check: t/τ is dimensionless.

E9: Ordinary exponential null limit

$$R_{exp}(t) = \exp(-t/\tau) = E_{1,1}(-t/\tau) \quad (18)$$

Dimensional check: t/τ is dimensionless.

E10: Stretched-exponential competitor

$$R_{SE}(t) = \exp[-(t/\tau)^\beta] \quad (19)$$

Dimensional check: t/τ is dimensionless.

E11: Bi-exponential competitor

$$R_{2e}(t) = we^{-t/\tau_1} + (1-w)e^{-t/\tau_2} \quad (20)$$

Dimensional check: Each exponential argument is dimensionless and w is dimensionless.

E12: Phase-resolution convergence diagnostic

$$\delta_c = \|\mathbf{c}_f - \mathbf{c}_c\|_2 / \max(\|\mathbf{c}_f\|_2, 10^{-30}), \quad \mathbf{c} = (a, b) \quad (21)$$

Dimensional check: Coefficient vectors are compared componentwise under the same coordinate units; ratio is dimensionless.

E13: Memory-resolution convergence diagnostics

$$\delta_\alpha = |\alpha_f - \alpha_c|, \quad \delta_\tau = |\tau_f - \tau_c| / \max(|\tau_f|, 10^{-30}) \quad (22)$$

Dimensional check: Both diagnostics are dimensionless.

Registered code-to-mathematics steps

1. **S1** [G0; equations E1]: Verify frozen protocol/blind commitments and enumerate hashed provider cases.
2. **S2** [G1; equations E1]: Validate provider metadata, array shapes, monotonicity, finite values and sampling uniformity.
3. **S3** [G2; equations E4, E9]: Evaluate Python FP64 finite differences, POD spectral stability and the Mittag-Leffler $\alpha=1$ exponential identity.
4. **S4** [G3; equations E1, E2, E3, E4, E5, E8, E9, E10, E11]: Fit Wave/Klein-Gordon/Duffing/Sine-Gordon and exponential/stretched/bi-exponential/Mittag-Leffler families on discovery blocks and evaluate BIC.
5. **S5** [G4; equations E6]: Evaluate frozen held-out temporal blocks with NRMSE ratios.
6. **S6** [G5; equations E12, E13]: Apply deterministic coarsening diagnostics and strict C++/OpenMP FP64 parity to confirmed candidates.
7. **S7** [G6; equations E12]: Consume optional canonical-framework SYCL/DD32 smoke evidence as screening-only diagnostics.
8. **S8** [G7; equations E5, E6]: Assess evidence-class-aware replication; synthetic/simulation controls cannot close physical cross-source replication.
9. **S9** [G8; equations E4, E8]: Classify joint, phase-only or memory-only physical support only after eligible replication.

8 Sources and provider contract

Real inputs satisfy A056-DYNAMIC-PROVIDER-4. Static knot geometry is not a substitute for $\varphi(t, s)$. Each physical case must carry a frozen phase definition, perturbation ID, solver/provider version, carrier and upstream geometry hashes, evidence class, independence unit and provenance family. The bundled synthetic cases are implementation controls only.

9 Gates

G0 frozen protocol and blind commitment;

G1 provider/admissibility;

G2 Python-FP64 analytic and spectral qualification;

G3 blind discovery;

G4 held-out confirmation;

G5 deterministic resolution checks and C++/OpenMP FP64 certification;

G6 optional SYCL FP32/DD32 screening, non-blocking for physical replication;

G7 evidence-class-aware physical cross-source replication;

G8 joint mechanism classification;

G9 commitment-verified reveal.

Synthetic/simulation evidence cannot close G7.

10 Numerics and convergence

Phase support requires positive Sine–Gordon coefficients, condition number $\leq 10^8$, discovery $\Delta\text{BIC} \geq 10$, and the registered held-out NRMSE-ratio threshold. Memory support requires the same BIC/holdout logic plus $0.48 < \alpha < 0.97$.

For $\mathbf{c} = (a, b)$,

$$\delta_c = \frac{\|\mathbf{c}_f - \mathbf{c}_c\|_2}{\max(\|\mathbf{c}_f\|_2, 10^{-30})}.$$

For memory,

$$\delta_\alpha = |\alpha_f - \alpha_c|, \quad \delta_\tau = \frac{|\tau_f - \tau_c|}{\max(|\tau_f|, 10^{-30})}.$$

Production profiles additionally require A056 C++ FP64 parity $\leq 10^{-10}$.

11 Backend authority

Lane	Precision	Authority
Python/NumPy	FP64	REFERENCE
C++17/OpenMP	FP64	CERTIFICATION
SYCL	FP32	SCREENING_ONLY
SYCL DD32 / FP32x2	two FP32 components	SCREENING_ONLY

DD32 is not IEEE FP64; its accuracy is measured against the FP64 reference. The canonical v1.0.4 external worker and oneAPI link/provenance policy are used unchanged.

12 Blindness and reveal

The framework creates nonced commitments for the private forbidden-term list and reveal payload. Blind source/output packaging excludes private/revealed content. Reveal is explicit and allowed only after a frozen FULL/CERTIFY campaign satisfying the configured reveal preconditions.

13 Statistics and model selection

Discovery uses

$$\text{BIC} = n \ln(\text{RSS}/n) + k \ln n.$$

Confirmation uses normalized holdout error. Sine-Gordon and Mittag-Leffler must each improve on the best registered competitor, not merely fit acceptably in isolation. No threshold is tuned after reveal.

14 Results

Implementation conclusion JOINT_CONTROL_RECOVERED

Physical conclusion NOT_ESTABLISHED

Certified cases 4 / 5

Case	Evidence	Phase winner	Relaxation winner	Certified
C0001	synthetic_control	SG	ML	True
C0002	synthetic_control	SG	BIEXP	True
C0003	synthetic_control	KG	ML	True
C0004	synthetic_control	KG	BIEXP	False
C0005	synthetic_control	SG	ML	True

Automatic run provenance

Field	Recorded value
Protocol bundle SHA-256	37fba708c58fbc7edc2068dd06157928841c3b09cb5280f4506faa29b9e14426
Framework schema	SST-FALSIFIER-PROTOCOL-2
Framework version	1.0.4
Python	3.14.3 (tags/v3.14.3:323c59a, Feb 3 2026, 16:04:56) [MSC v.1944 64 bit (AMD64)]
Platform	Windows-11-10.0.26300-SP0

Gate ledger

Gate	Status	Question	Reason
G0	PASS	Is the frozen A056 protocol intact and is the blind boundary verifiable?	Framework verified frozen protocol; A056 blind commitment also verified.

G1	PASS	Are all runtime provider cases present, hashed, contract-valid and numerically admissible?	
G2	PASS	Do Python/NumPy FP64 reference numerics pass analytic sanity checks and spectral qualification?	
G3	PASS	Does blind discovery identify preregistered Sine-Gordon and/or Mittag-Leffler candidates against the competitor families?	
G4	PASS	Do discovered candidates survive the preregistered held-out temporal confirmation blocks?	
G5	PASS	Do confirmed candidates survive resolution checks and strict C++/OpenMP FP64 parity against the Python FP64 reference?	C++ FP64 parity executed; case-level resolution failures remain excluded from replication.
G6	DEFERRED	If explicitly run, does optional SYCL FP32/DD32 screening satisfy the frozen framework parity contract without being promoted to certification?	No instance-local SYCL/DD32 smoke supplied; screening is optional and non-blocking.
G7	NOT_RUN_PREREQUISITE	Does certified support replicate across eligible physical evidence classes without pseudo-replication?	Only implementation-control/simulation evidence is available; physical cross-source replication is ineligible.
G8	NOT_RUN_PREREQUISITE	Does physically replicated evidence close the preregistered joint Sine-Gordon plus fractional-memory mechanism claim?	Physical cross-source replication did not PASS.
G9	PASS	After the blind phase, do the reveal and blind-term commitments verify exactly?	Reveal prerequisites and both nonced commitments verified.

Backend summary

Field	Recorded value
A056 version	v0.2.3
Framework	1.0.4
Python reference	python-numpy-fp64 / float64 / REFERENCE
C++ certification	PASS / CERTIFICATION
Native backend	openmp
Compiler	C:\Program Files\Microsoft Visual Studio\2022\Community\VC\Tools\MSVC\14.44.35207\bin\Hostx86\x64\cl.exe
Compiler version	Microsoft (R) C/C++ Optimizing Compiler Version 19.44.35228 for x64
Build fingerprint SHA-256	ffef43bde495e9ed28c3266275fc65a1cf94b04658da023ff68f8545d0d7da28
GPU screening	DEFERRED / SCREENING_ONLY

Source summary

Source	Role	Evidence	Provenance family
A056_SYNTH_ CONTROLS	implementation_ control	synthetic_control	A056_SYNTHETIC_ GENERATOR_V01

Full machine-readable provenance is preserved in `BACKEND_MANIFEST.json`, `SOURCE_MANIFEST.json`, `ENVIRONMENT_PUBLIC.json`, and `OUTPUT_MANIFEST.json`.

15 Interpretation

Synthetic-control recovery validates implementation/model-selection behavior only. Simulation-only support remains simulation evidence. Physical support requires G5 certification and G7 eligible replication. A joint SST mechanism claim requires both phase and memory closure to replicate jointly and close G8.

16 Reproducibility

The frozen protocol bundle hashes `falsifier.toml`, `science/source/gate/blind` contracts, both commitment files and this report. Run outputs record environment, source, backend, gate and output manifests. The framework is pinned in `FRAMEWORK_PIN.json`; A056’s instance-local C++ source is hash-bound by the canonical framework build provenance.

References

- [1] A. C. Scott, F. Y. F. Chu, and D. W. McLaughlin, “The soliton: A new concept in applied science,” *Proceedings of the IEEE* **61** (1973), 1443–1483. [doi:10.1109/PROC.1973.9296](https://doi.org/10.1109/PROC.1973.9296).
- [2] I. Podlubny, *Fractional Differential Equations*, Academic Press, 1999.
- [3] R. Gorenflo, A. A. Kilbas, F. Mainardi, and S. V. Rogosin, *Mittag-Leffler Functions, Related Topics and Applications*, Springer, 2014. [doi:10.1007/978-3-662-43930-2](https://doi.org/10.1007/978-3-662-43930-2).
- [4] I. T. Jolliffe, *Principal Component Analysis*, 2nd ed., Springer, 2002. [doi:10.1007/b98835](https://doi.org/10.1007/b98835).
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